



How to Treat Compulsive Facets of Behavioural Addictions

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Abstract

Purpose of Review Treatment approaches to disorders of behavioural addiction are beginning to emerge. In this paper we provide a narrative review of the growing range of treatment approaches for the less well understood compulsive facets, including compulsions and the associated latent phenotypes, such as cognitive domains characterised by inflexible thinking or a preference for habitual responding. We aim to offer new insights and heuristics for treating compulsive forms of behavioural addiction, such as cyberchondria, compulsive hoarding, compulsive forms of gaming disorder, trichotillomania and skin picking (excoriation) disorder.

Recent Findings Emerging evidence of treatment approaches showing promise include novel forms of pharmacotherapy (extending beyond drugs acting on the serotonin system), psychotherapy (extending beyond conventional cognitive-behaviour therapy) and non-invasive neuromodulation.

Summary A broad approach to designing novel treatment strategies that includes careful characterisation of the phenotypic and latent cognitive profiles based on a growing understanding of the neurobiology of compulsivity is recommended.

Keywords Treatment approaches · Behavioural addiction · Compulsivity · Pharmacotherapy · Psychotherapy · Non-invasive neuromodulation

Introduction

Disorders due to addictive behaviours comprise a newly recognized category of clinically significant syndromes associated with distress or interference with personal functions that develop because of loss of control of rewarding behaviours other than the use of dependence-producing substances. The Diagnostic and Statistical Manual Version 5 (DSM-5) recognizes gambling disorder in the category of ‘Substance-Related and Addictive Disorders’ [1] – this being the first behavioural addiction to be formally recognized as such by the American Psychiatric Association (APA). The International Statistical Classification of Diseases and Related Health Problems 11th edition [2] names gaming disorder and gambling disorder as the first specified disorders due to addictive behaviours. The boundaries of behavioural addiction remain somewhat ill-defined, owing at least in part to relative lack of data. Thus, behaviours that are currently listed in other ICD-11 categories, such as compulsive sexual behaviour disorder (which includes pornography use), skin picking disorder and trichotillomania, may be alternatively conceptualised as behavioural addictions as the constituent urge-driven repetitive behaviours often

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manifest reward-seeking features. Other forms of addictive behaviour not currently listed in the DSM-5 or ICD-11 frequently take place online and include loss of control of buying or shopping, pornography use, social media use, cyberchondria (online searching for illness related information), digital hoarding, cyberstalking, and excessive use of online streaming [3].

Two main subtypes of behavioural addictions have been proposed, based on the underlying motivational state: one seemingly related to poor impulse control (i.e., gaming, gambling, buying/shopping, cybersex/pornography use, social media use, kleptomania, food addiction) and another, seemingly related more strongly to compulsive tendencies [3, 4] (i.e., cyberchondria, digital hoarding and possibly some forms of gaming as well as hair pulling or skin picking). Impulsive behaviours are actions that are inappropriate, premature, inadequately thought out and risky, primarily motivated by reward (positive reinforcement) [5]. In contrast, compulsions are repetitive, stereotyped precautionary actions, repeated despite adverse consequences and designed, at least initially, to avert distress or adverse consequences (negative reinforcement) [6, 7]. There is substantial overlap, however, with the experience of loss of control of the urge to act being central to both impulsive and compulsive behaviours. Behaviours that initially appear impulsive may become compulsive over time and it is thought that impulsivity and compulsivity are driven by at least some common neurobiological processes, such as lack of ‘top-down’ cortical executive control, or ‘disinhibition’ [8]. Differentiation of behavioural addictions into more impulsive and compulsive forms is likely to have clinical relevance in terms of indicating dominance of different neuropsychological mechanisms and by extension efficacy of different treatment approaches.

The main established treatment modalities for behavioural addictions comprise various forms of psychological therapy and to a lesser extent pharmacotherapy; however, there is insufficient evidence to determine which of these modalities shows greater efficacy, or whether combined approaches confer added benefit. Most interventional research has focused on gambling disorder and more recently gaming disorder; and there has been little research into prevention or treatment of compulsive forms of behavioural addictions, such as cyberchondria, for which arguably no ‘evidence-based’ treatments exist.

In this paper, we review treatment approaches for the less well understood *compulsive facets*, to offer new insights and heuristics for treating compulsive forms of behavioural addiction. The general difference between the treatment of addiction and obsessive–compulsive disorders, that we draw upon in this chapter, is a greater emphasis on biological forms of treatment such as pharmacotherapy and neurostimulation for the latter. We consider the compulsive facets

of behavioural addiction both as the observed phenotype, i.e., compulsions (see below), and the latent (hidden) phenotypes thought to underpin compulsions, including cognitive domains characterised by inflexible thinking i.e., the inability to flexibly shift responding when contingencies change, or a preference for habitual responding i.e., the tendency to repeat actions even though they are no longer purposeful, rewarding or necessary [9].

Pharmacological Treatment of Compulsions

As the archetypal compulsive disorder, obsessive–compulsive disorder (OCD) has been relatively well researched. Obsessions in OCD are recurrent, intrusive, highly distressing, thoughts, often involving personalised harm-related themes (e.g., I may be responsible for an accident), which are usually accompanied by compulsions, which are unwanted, stereotyped, urge-driven repetitive acts (e.g., checking, washing, ordering), almost always motivated by a drive to reduce harmful consequences, at least at the outset of illness. Evidence based treatments for compulsions in OCD include forms of cognitive behaviour therapy (CBT) and pharmacotherapy. The established pharmacotherapy of OCD is relatively circumscribed and includes serotonin reuptake inhibitors (SRIs), such as the selective serotonin reuptake inhibitors (SSRIs) or clomipramine, with greater benefits derived from dosages higher than those used for depression. SRIs are also effective at preventing relapse, which is important for chronic disorders such as OCD and behavioural addictions. For those with OCD responding poorly to SRI, adjunctive dopamine antagonists are effective in some cases, typically effective at doses lower than would be used for other indications (e.g., psychosis) [10].

Other promising pharmacological interventions for those with OCD failing to respond to SRI include drugs thought to regulate excitatory intracerebral glutamate activity (e.g., memantine, N-acetyl-cysteine [N-AC], l-carnosine), or those acting as voltage sensitive neuronal membrane channel blockers (e.g., lamotrigine, topiramate), as well as drugs increasing intracerebral dopamine transmission such as d-amphetamine [11] and tolcapone [12]. There is also some evidence that morphine may be helpful [13], hinting at a role for opiate mechanisms, while the opiate antagonist naltrexone was not effective in OCD, making patients more anxious and depressed [14].

Compulsions associated with some other ICD-11 obsessive–compulsive and related disorders, such as hypochondriasis, appear phenotypically similar to those of OCD. Randomised controlled trials (RCTs) show hypochondriasis responds to SRIs rather like OCD [15], though there is insufficient evidence to suggest other ‘experimental’ medications for OCD, reviewed above, are effective. As far as the authors are aware, there have been no RCTs of pharmacological

treatment for cyberchondria, which resembles hypochondriasis [16]. It might be expected however that SRIs are potentially efficacious.

In contrast, the compulsions of trichotillomania and skin-picking disorder seem phenotypically different from OCD, as they are often linked to a sense of accomplishment or transient reward (i.e., getting rid of a kinky hair that does not belong or a bump of the skin that is out of place) [17]. RCTs of SSRIs have produced disappointing results in these disorders (though positive studies of clomipramine in trichotillomania and SSRI in skin picking disorder exist), suggesting that these body-focused compulsions respond differently from OCD. In trichotillomania, RCT evidence suggests that monotherapy with the dopamine antagonist olanzapine [18] and the glutamatergic agents N-AC [19] and memantine [20] are effective, while in skin-picking disorder, N-AC [21], memantine [20], and naltrexone [22] have been shown to be efficacious, the latter hinting that brain opiate systems may be relevant for treating skin-picking disorder.

Hoarding compulsions are highly stigmatising and treatment refractory, predicting a poor response to SRI in OCD [23, 24]. Their treatment is under-researched, but an open label study of hoarding disorder (in the absence of OCD) suggests possible benefit may be derived from venlafaxine [25]. No effective pharmacological treatment for digital hoarding has yet been identified [26].

Pharmacology of Cognitive Flexibility and Habits

Precision psychiatry aims to tailor pharmacological treatments to meet patients' individual needs rather than being disorder specific. This goal is consistent with the Research Domain Criteria (RDoC) from the National Institute of Health in the United States which, in part, looks to transdiagnostic elements such as cognitive domains as potential treatment targets. One such cognitive domain is cognitive flexibility, impairment of which is associated with core aspects of several psychiatric disorders, such as compulsivity (the feeling that things need to be done in the same way repetitively as in OCD, eating disorders, addictive disorders) and habit (doing the same things over and over again, even without much thought, as in trichotillomania, skin picking, behaviours associated with autism spectrum disorders).

In the RDoC matrix, cognitive flexibility is probably best considered under the construct umbrella category of 'cognitive control', which includes flexible updating and goal selection/performance monitoring [27]. Inflexible responding can contribute to excessive habit, since due to inflexibility an action can be repeated despite being functionally impairing or deleterious to long-term goals. How best to examine, subdivide and conceptualise cognitive flexibility remains open to debate, including in relation to other cognitive domains. Elegant work over many years has shed light

on neurochemical systems involved in different aspects of flexible responding, considered in detail elsewhere [28–30]. Here, we focus on examples of pharmacological manipulation of flexible responding as pertain to models of psychiatric symptoms, focusing on two common cognitive models/tasks.

Cognitive flexibility is vital for adaptive behaviour, and is broadly defined as the ability to switch between different cognitive operations in response to changing environmental demands. Cognitive flexibility is typically measured as perseveration on prior rules in set-shifting or rule switching tasks. Tasks measuring cognitive flexibility include the Wisconsin Card Sorting Test (WCST) and the Cambridge Neuropsychological Test Automated Battery (CANTAB) Intra-Extra Dimensional Set Shift (ID/ED) test. Both are sensitive to disruption of fronto-striatal circuitry [28]. Importantly, work using the ID/ED and its variants has demonstrated double dissociations whereby the dorsolateral prefrontal cortex (DLPFC) is important to extra-dimensional shifting, and the orbitofrontal cortex (OFC) is important to reversal learning [28, 31, 32].

Animal research sheds light on how we might improve cognitive flexibility in certain models of psychiatric symptoms. While animal models cannot capture the full gamut of symptoms evident in compulsive conditions, they have potential advantages when attempting to understand the neurocircuitry and neurochemical pathways implicated in particular cognitive processes. For example, animal models can encapsulate specific genetic, neurochemical, and/or environmental manipulations, and can enable access to brain tissue measurement *in vivo*, which is not feasible or appropriate in humans (for further discussion see e.g. [27, 33–35]). Animal models can thus shed light on implicated pathophysiological mechanisms relevant to human disease, which may in turn signal new treatment approaches or help to understand mechanisms underlying efficacy of existing treatments. At the same time, it is important to bear in mind the limitations of animal models—for example, animal models can lack validity versus human disease in terms of face validity, construct validity, and/or predictive validity [36]. Another limitation can be the over-interpretation of animal models: for example, models exist in which rodents excessively groom their cage mates; such models have been purported to represent trichotillomania or even OCD. However, the core phenomenology in trichotillomania is pulling out of one's own hair, thus these particular animal models have limited phenomenological similarity to trichotillomania, let alone OCD which involves not only compulsions but also intrusive thoughts on particular themes.

Autism spectrum disorders are associated with deficits in social interaction and communication, repetitive stereotyped behaviours, and restricted interests (i.e., cognitive rigidity). One study examined, in a rodent model of core

autism-related phenotypes, the acetylcholinesterase inhibitor donepezil. Donepezil significantly decreased cognitive rigidity, improved social preference, and enhanced social interaction in a dose-dependent manner when injected into the dorsomedial striatum [37]. Another animal study relevant to autism spectrum disorders examined afobazole, a non-benzodiazepine anxiolytic drug that acts via activation of Sigma1R and interacts with melatonin receptors and a regulatory site of MAO-A. The study found that, in a rodent model of autism spectrum disorders, afobazole promoted retraining and reversal learning [38].

In humans, cognitive inflexibility is common in people with eating disorders who often present with inflexible eating-related behaviours. In a study of intranasal oxytocin, 41 patients with anorexia nervosa were randomized to oxytocin or placebo. Those receiving intranasal oxytocin performed significantly better on the WCST after 4 weeks of treatment, suggesting oxytocin may improve cognitive flexibility [39].

The most common human psychiatric disorder associated with cognitive inflexibility is probably OCD [40]. Intuitively, it appears logical that the compulsions of OCD could result from cognitive inflexibility, since compulsions are repetitive and the person feels compelled to engage in the behaviours according to rigid rules and/or due to obsessive thoughts. Studies examining pharmacological treatment for OCD have examined both obsessions and compulsions but, when both improve, it is difficult to parse which the medication is modifying and how this may link to flexibility. Intriguingly, some recent data suggest that SSRIs do not improve, and may even worsen, cognitive flexibility [41, 42]. Thus, how SSRIs improve OCD may not be via cognitive flexibility, as operationalized by cognitive tasks.

In a double-blinded RCT of 36 people with OCD who were assigned topiramate or placebo [43], topiramate had a significant effect on compulsions but not obsessions, suggesting that modifications in glutamatergic function may be responsible, at least in part, for the improved response in compulsions. This is the only study we are aware of that found medication improved compulsions and not obsessions. The findings suggest that the constructs are dissociable and some medications could target compulsions without first affecting obsessions; but that the link between cognitive inflexibility and compulsions is not clear cut.

In skin picking disorder, a small placebo-controlled RCT found that lamotrigine, a medication that targets medial prefrontal glutamatergic drive to the nucleus accumbens, conferred benefit only in a subset of patients who exhibited relatively impaired cognitive flexibility on the ID/ED task [44]. In gambling disorder, an open label study examined memantine, an N-methyl D-aspartate receptor antagonist, demonstrating that it was associated with reduced gambling and improved ID/ED performance [45]. These findings

suggested that pharmacological manipulation of the glutamate system may target cognitive inflexibility.

More recently, the question has arisen as to whether psychedelic pharmacological agents may target cognitive flexibility. Activation of 5-HT_{2A} receptors with psychedelic drugs has been found to acutely impair cognitive flexibility in humans [46], while the effects of their activity at the 5-HT_{2C} receptor on cognitive flexibility are not well understood [47]. However, psilocybin has shown tentative early potential promise for treating certain cases of mood disorders, which are often accompanied by cognitive dysfunction including inflexibility [48]. An open-label study of 24 patients with major depressive disorder found that psilocybin may modulate perseverative errors on a set-shifting task and neurometabolite concentrations in brain regions supporting cognitive flexibility (e.g., the anterior cingulate cortex; ACC) [49]. Psilocybin was associated with increased cognitive flexibility for at least 4 weeks post-treatment, though these improvements were not correlated with the antidepressant effects and the absence of a control for the psilocybin means any effects cannot be attributed with certainty to the medication.

Overall, then, emerging evidence suggests that modulations of particular neurochemical systems, including acetylcholine, oxytocin, glutamate and serotonin, can impact cognitive flexibility, quantified using cognitive tasks, for some psychiatric symptoms linked to habit. This evidence includes pre-clinical models as well as studies in humans with specific mental disorders. Controlled clinical trials are needed for habit-related mental disorders that include such cognitive tasks, used alongside gold-standard clinical outcomes. Such research may not only reveal novel trans-diagnostic approaches but also shed light on mechanisms underlying treatment response and account for individual variation in treatment efficacy.

Non-invasive Neurostimulation for Compulsions, Cognitive Inflexibility and Habits

Non-invasive neurostimulation is an umbrella term for various methods of modulating electrical activity in cortical nodes and underlying neurocircuitry, using techniques such as repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (dTCS). Non-invasive neurostimulation protocols show promise in reducing addictive behaviours and cravings associated with nicotine smoking, alcohol use disorder, cocaine addiction, and food cravings [50]. The techniques may induce changes in behaviour and neural circuitry by inducing cortical plasticity [51]; however, the underlying processes leading to putative therapeutic benefits are not fully understood.

In depression, rTMS on the left DLPFC demonstrated improvement in symptoms and cognitive measures related to

disinhibition and inflexibility versus sham [52]. Deep TMS and rTMS targeting the ACC in patients with OCD alleviated symptoms versus sham and influenced cognitive control markers, such as the Error-Related Negativity, measured using electroencephalographic recordings [53, 54].

Although heterogeneous in protocols and parameters, results from rTMS studies converge in indicating that the stimulation (instead of inhibition) of prefrontal regions could be beneficial to counteract dysfunctional gambling-related decision processes [55]. In a recent meta-analysis of rTMS for gambling disorder [56] (8 studies, of which 3 were RCTs (Table 1)), excitatory rTMS of the left PFC showed some effects on measures potentially relevant to gambling (e.g., craving); however further standardization of stimulation protocols and psychometric assessments is needed. Future studies should also use valid outcome measures for quantifying effects of intervention on gambling disorder symptoms, which these studies generally did not do.

TMS also holds promise as a potential treatment for other behavioural addictions [57, 58]. In a single open-label study of patients with trichotillomania, hoarding disorder, and skin picking disorder [59], rTMS applied over the left DLPFC produced a short-lived symptom reduction in all three groups, with sustained benefits only in younger patients, suggesting that the stability of TMS effects decreases with age, possibly due to a decline in brain plasticity. However, controlled data are lacking.

Moving to tDCS, a meta-analysis in OCD [60] including eight studies showed a significant effect of treatment at study endpoint but active tDCS was not superior to sham when only RCTs were included. In specific RCTs, tDCS marginally reduced ERN amplitude (a measure of performance monitoring) [61], and open-label tDCS administered across the OFC was associated with improved symptoms while also potentially changing the functioning of GABA-B receptors [62]. An unpublished secondary analysis of a small feasibility study of tDCS in OCD [63] showed that stimulating the OFC and supplementary motor area (SMA) with a short course of tDCS significantly improved cognitive flexibility versus sham.

In gambling disorder, tDCS targeting the DLPFC [64] improved performance on a decision-making task and cognitive flexibility in the WCST versus sham. In a study on tDCS versus sham in children with attention deficit hyperactivity disorder [65], inhibitory control (measured via a go/no-go task) was improved by tDCS applied to the DLPFC, while WCST performance was improved by combined DLPFC-OFC stimulation, but not DLPFC stimulation alone.

Altogether, these findings provide preliminary evidence suggesting targeting cortical regions associated with cognitive control and flexibility using non-invasive neurostimulation may be helpful for addressing compulsive symptoms and disorders, of great relevance to the study and treatment

of behavioural addiction. However, such approaches remain exploratory until large rigorous controlled trials are conducted to clarify efficacy and tolerability. Attempts have been made to achieve greater clarity about the relative efficacy of these different targets and protocols by pooling data and applying meta-analysis. In regard to TMS studies, earlier meta-analyses identified the OFC [66] as the most efficacious target, but, as studies accrued, later meta-analyses favored first the SMA, and then the bilateral DLPFC as the possible most effective target [67]. However, the high levels of heterogeneity in clinical outcomes seen across all the previous meta-analyses, and thought to be associated with major differences in individual trial design and methodology, has continued to undermine confidence in the specific brain areas that could be the most effective to stimulate. For studies on tDCS, the OFC area seems to hold promise [63], but more data is needed to corroborate this finding.

Psychotherapeutic Treatments of Compulsivity in Behavioural Addictions

The psychotherapeutic management of compulsivity in behavioural addictions aims to address the underlying psychological vulnerabilities that drive/maintain compulsive behaviours, help improve self-control, and develop/maintain healthier coping strategies. Examples of the considered psychotherapeutic approaches in the treatment of behavioural addictions include cognitive behavioural therapy (CBT), exposure and response prevention (ERP) and habit reversal therapy (HRT) (Table 2) among others.

Cognitive Behavioural Therapy

Several studies have highlighted the potential effectiveness of CBT in treating gambling disorder. A number of RCTs and meta-analysis [68], have found sustained benefit, including reductions in gambling behaviour and improved quality of life. However, in many of the studies the control arm was inadequate e.g., involved a waiting list. In a study that compared CBT with motivational interviewing (MI) [69], whereas both CBT and MI reduced gambling behaviours, CBT was more effective for addressing cognitive distortions associated with gambling. Some researchers consider CBT delivered on its own may not always be the optimal approach. Ingle et al. [70] explored the effectiveness of integrating CBT with other treatment modalities, such as family therapy, finding that a multi-modal approach could enhance outcomes.

CBT was adapted and tested in various studies for its effectiveness in gaming disorder, focusing on the maladaptive thoughts and behaviours that contribute to the compulsive use of online games [71]. A systematic review and CONSORT evaluation [72] assessing clinical trials on

Table 1 Randomised Controlled Trials of repetitive transcranial magnetic stimulation (rTMS) in gambling disorder

Author (year)	Country	Study Design	Setting	Population	Intervention	Stimulation protocol	Comparator	Gambling-related outcomes	Other outcomes	Timeframe for follow-up	Results
Gay et al. (2017)	France	RCT	Not specified	N = 22 (14 males—8 females)	rTMS	Sessions: single session 110 of RMT; 10 Hz; 94 trains of 3.2-s duration at 10-s interval, for a total of 3008 pulses per session and total treatment duration of 20 min 30 ss	TMS-Sham	-Yale-Brown-Obsessive-Compulsive Scale adapted for Pathological Gambling (PG-YBOCS) 100-mm visual analogue scale	None	Baseline	Cue-induced craving (VAS): improvement
Zack et al. (2016)	USA	RCT	Others (community recruited)	N = 9 (9 males)	rTMS	rTMS, single session	TMS-Sham	Desire to gamble (VAS)	Stroop task	Before and after each stimulation	Desire to gamble (VAS)
				Mean Age (SD): 43.2 (13.2)	High frequency	Stimulation parameters: 80% of the AMT; 10 Hz; Three separate epochs of rTMS were administered with a 5-min break between epochs. For each epoch, 15 sets of 10 pulses were delivered with a frequency of 10 Hz, and a 10-s break was given between sets. Total pulses 450		Numeric scale for cue-induced craving	-Delayed discontinuing task		rTMS: Improvement in comparison to sham
				Diagnosis: DSM-IV Criteria	Area: left DLPFC	cTBS, single session		Numeric scale for desire to gamble; No improvement			Numeric Scale for desire to gamble; No improvement
				Mean Age (SD): 51 years (13.7)	High Frequency	cTBS, single session		PG-YBOCS; No improvement		7 days	PG-YBOCS; No improvement
				Diagnosis: PD	Area: mPFC	cTBS, single session		-Blood pressure			-cTBS: No differences in comparison to sham
					cTBS Area: Right DLPFC	Stimulation parameters: 80% of AMT. Each TBS burst consisted of three pulses at 50 Hz, with each train being repeated every 200 ms (5 Hz). cTBS consisted of continuous repetition of trains for 20 s (300 pulses)		-Profile of Mood State (POMS)			-cTBS: Improvement in comparison to sham
								-Addiction Research Center Inventory (ARCI)			-rTMS: No differences in comparison to sham
											-cTBS: Improvement in comparison to sham
											-rTMS and cTBS: No differences in comparison to sham

Table 1 (continued)

Author (year)	Country	Study Design	Setting	Population	Intervention	Stimulation protocol	Comparator	Gambling-related outcomes	Other outcomes	Timeframe for follow-up	Results
Sauvaget et al. 2018	France	RCT	Out-patients	N = 30	rTMS	Sessions: single session	TMS-Sham	- Visual Analog Scale (VAS) for craving	-Heart rate (bpm)	-At baseline	- VAS-cue induced craving: No differences between active and sham rTMS
				Mean Age (age interval): Active arm: 33 (28–42)	Low frequency	Stimulation parameters: 120% of the RMT, 1 Hz with one train producing 360 pulses in a single 6-min session		-Gambling Craving Scale (GACS) (only 3 first questions)	-Systolic blood pressure	-Before the rTMS session	- GACS (gambling-related craving) -3 items used to measure the desire to gamble: No differences between active and sham rTMS
				Sham arm: 39 (34.5–56)	Area: right DLPFC				-Diastolic blood pressure	-Immediately after the rTMS session	-Heart rate: No differences between active and sham rTMS
				Diagnosis: 5 or more of the DSM-IV diagnostic criteria for GD						-Every 5 min until the craving intensity returned to the baseline level	-Systolic blood pressure: No differences between active and sham rTMS
											- Diastolic blood pressure: No differences between active and sham rTMS

DLPFC, dorsolateral prefrontal cortex; rTMS repetitive transcranial stimulation; NA, not applicable; RCT, randomized controlled trial; DSM-5, Disorders-Fifth edition; GD, gambling disease; RMT, resting motor threshold; cTBS, continuous theta burst stimulation; pre-SMA, pre-supplementary motor area; mPFC, medial prefrontal cortex; PD, Parkinson Disease; AMT, active motor threshold; DSM-IV Diagnostic and Statistic Manual of Mental Disorders-IV; TBS, theta burst stimulation; SD, standard deviation; NR, Not Reported

Table 2 Key Elements Of Cognitive Behaviour Therapy Approaches In Behavioural Addictions**A. Cognitive Behavioural Therapy**

1. Identify Triggers: Understand situations, emotional states, or thoughts that lead to compulsive behaviour
2. Cognitive Restructuring: Challenge and change the distorted beliefs and thought patterns that justify or perpetuate the compulsive behaviour
3. Skills Training: Develop alternative, healthy coping strategies for dealing with stress, emotional distress, or triggers
4. Problem-Solving Skills: Enhance the individual's ability to deal with life's challenges without resorting to compulsive behaviour
5. Relapse Prevention: Identify potential relapse triggers and create strategies to manage them without falling back into old patterns

B. Exposure and Response Prevention

1. Systematic exposure: In real life to cues or situations that trigger urges to perform compulsive behaviours
2. Response prevention: Prevent engagement in compulsive behaviour in order to sustain the urges, which are eventually extinguished

C. Habit Reversal Therapy

1. Awareness training: Enhance self-awareness of the cues and triggers leading to compulsive behaviours
2. Development of a competing response: Teach individuals to interrupt the compulsive behaviour with a competing non-pathological motor response, usually involving a similar muscle group
3. Social support: Enlist help from others to encourage the use of the technique

Internet addiction treatment found evidence supporting CBT's effectiveness though highlighted the need for more rigorously designed studies. A study [73] demonstrating that a mindfulness-oriented CBT was effective in gaming disorder hints that integrating CBT with other therapeutic techniques could enhance treatment outcomes. Although a systematic review [74] found that CBT-based interventions showed promising results in gaming disorder, a more recent meta-analysis [75] found only limited evidence suggesting that CBT is an effective intervention and argued for more rigorous studies with larger sample sizes, longer follow-up periods, and standardized measures to establish efficacy, identify long-term benefits and optimize treatment protocols.

Emerging evidence suggests compulsive mechanisms may play a key role in harmful online sexual behaviour. CBT may be an effective approach for some individuals, by addressing underlying cognitive and emotional drivers and helping individuals develop healthier coping strategies [76–80]. However, more research, including RCTs with long-term follow-up, is needed to establish efficacy, refine treatment protocols, and identify therapeutic mechanisms.

Exposure and Response Prevention (ERP)

ERP is a behavioural form of CBT known to be effective in OCD [81] and related disorders such as hypochondriasis [15] (Table 2). It is grounded in the principles of classical conditioning and is believed to facilitate learning that the expected negative consequences of not engaging in compulsive behaviour are either manageable or non-existent. Over time, this process reduces the urge to perform the behaviour and decreases the anxiety associated with cues. An alternative theoretical inhibitory learning model argues that ERP helps by creating new learning experiences that compete and can be used by the patient to override the old compulsive behaviours, as opposed to its effect being limited

to just extinguishing the old behaviours [82]. The use of ERP in behavioural addictions is less well-explored than in OCD and requires adaptation to accommodate the specific challenges of these disorders, including techniques for self-management of relapse prevention.

Some studies have adapted ERP protocols for gambling disorder, focusing on exposing individuals to gambling-related cues (such as images of slot machines or casino-related sounds) while preventing gambling responses. A study by Smith and colleagues [83] demonstrated that this approach could reduce gambling urges and behaviours. More extensive and robust empirical support is required to address ERP treatment protocols for other specific behavioural addictions, gain improved understanding of long-term outcomes, and understand its therapeutic mechanisms. Examination of ERP augmentation with pharmacological interventions and/or other psychotherapeutic modalities (e.g., HRT) should also be considered.

Habit Reversal Therapy (HRT)

HRT (Table 2) was primarily developed for treating tic disorders and body-focused repetitive behaviours such as trichotillomania and skin picking disorder. It has also been suggested for alcoholism and binge purging behaviors and so may be considered for behavioural addictions [84]. This method of CBT aims to stop unwanted behaviour and replace it with a more adaptive coping mechanism, thereby addressing the underlying compulsivity that drives behavioural addiction [84].

The utility of HRT in treating disorders underpinned by compulsive behaviours, such as trichotillomania and skin picking disorder, is supported by a limited amount of clinical research [85]. Patients with skin picking disorder treated with HRT exhibited sustained improvement over a 12-month follow-up [86], suggesting HRT could play a long-term role.

In gaming disorder, HRT combined with CBT produced significant improvements in gaming-related compulsivity, time spent gaming, and overall psychological well-being [87]. Another comparative study investigating the effectiveness of HRT versus CBT in treating internet addiction, found that while both approaches significantly reduce compulsive internet use, HRT showed promising results in addressing specific compulsive behaviours [88].

Thus, while existing evidence supports the use of HRT in treating compulsivity in behavioural addictions, major limitations exist. These include a need for larger scale RCTs in a wider range of specific behavioural addictions, as well as studies examining its therapeutic mechanisms, treatment protocol optimization, and evaluation of its effectiveness in conjunction with other treatment modalities.

Potential New Approaches in Treating Compulsive Facets of Behavioural Addictions

As compulsive behaviours can be phenomenologically distinct and respond to different interventions, future approaches may benefit from careful characterisation of latent phenotypes (e.g., the cognitive domains underlying patients' compulsive behaviours). For example, computational analysis of probabilistic reversal learning in stimulant use disorder and OCD found that the groups differed in opposite directions in a “stimulus stickiness” parameter measuring the extent to which a stimulus is persistently chosen regardless of outcome [89]. The $D_{2/3}$ antagonist amisulpride reduced the elevated stickiness found in the SUD group but did not affect the reduced stickiness seen in the OCD group, suggesting that such latent phenotypes could potentially be used to better target existing treatments.

In tandem with this approach, novel treatment strategies are required. In terms of new pharmacological targets, a recent study in nonhuman primates demonstrated that a positive allosteric modulator at the M_1 muscarinic acetylcholine receptor (mAChR) reduced perseverative responding in an ID/ED task [90, 91]. Another avenue for further exploration will be new treatment combinations, such as combining neurostimulation with ERP, which has been shown to reduce compulsive behaviour in a rat model [92, 93].

Conclusions

Compulsive symptoms and mechanisms are core to many forms of behavioural addiction. However, there is a range of phenomenologically distinct compulsive behaviours and latent phenotypes that may require different forms of interventions. While emerging evidence suggests that differentiating patients with behavioural addictions according to presence of compulsive facets may have clinical utility, and that some

pharmacological, neurostimulation and psychological interventions with evidence of efficacy in ‘archetypal’ compulsive disorders, such as OCD, show some promise in this subgroup, a direct correlation does not exist and many such interventions, such as SSRI, do not seem to work in the majority of patients. A broad approach to designing novel treatment strategies that includes careful characterisation of the phenotypic and latent cognitive profiles based on a growing understanding of the neurobiology of compulsivity is recommended.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of Interest SRC receives honoraria from Elsevier for journal editor work. NAF has recently held research or networking grants from the NIHR, COST Action, Orchard, Horizon Europe, UKRI; accepted travel and/or hospitality expenses from the BAP, ECNP, RCPsych, CINP, World Psychiatric Association; received payment from Elsevier for editorial duties and the Mental Health Academy and Children and Screens for lecturing. She leads an NHS treatment service for OCD. She holds Board membership for various registered charities linked to OCD. She gives expert advice on psychopharmacology to the UK MHRA. JEG has received research grants from Janssen and Biohaven Pharmaceuticals. He receives yearly compensation from Springer Publishing for acting as Editor-in-Chief of the Journal of Gambling Studies and has received royalties from Oxford University Press, American Psychiatric Publishing, Inc., Norton Press, and McGraw Hill. The views expressed in this publication are those of the author(s) and not necessarily those of the NIHR, NHS or the UK Department of Health and Social Care.

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